

Statistics

Evaluation System

Percentage of In-Term Studies: 60%

1 Midterm Exam: 25%

2 Midterm Exam: 20%

Homework(s): 15%

Percentage of Final Exam: 40%

* It is obligatory to attend the courses for the students who have registered for the first time or whose last grade is F0.

Recommended Sources

1. Probability and Statistics for Engineering and Scientists, R.E. Walpole, R.H. Myers, S.L. Myers, K. Ye, Prentice Hall, 2007.
2. John E. Freund's Mathematical Statistics, Miller I., Miller M, Prentice Hall, 6th edition, 1999.
3. P.L. Meyer, "Introductory Probability and Statistical Applications", Addison-Wesley Pub., 1978.

Course Learning Outcomes

1. Displaying data through tables and graphics
2. Interpreting data from these tables and graphics
3. Summarizing data by using descriptive statistics tools
4. Having knowledge about basic concepts of sampling theory
5. Using estimation theory
6. Making decisions by using hypothesis testing
7. To be able to make and interpret regression analysis

Weekly Subjects and Related Preparation Studies

Week Subjects

- 1 Definition and Basic Concepts of Statistics, Graphical Projection of Data, Frequency, Diagrams and Distributions, Parameters
- 2 Means, Deviations, Skewness Coefficients
- 3 Quartiles and Box Diagrams, Sampling Theory
- 4 Normal Distribution, t- Distribution, Chi-squared Distribution, F Distribution, Central Limit Theorem, Sampling Distribution
- 5 Estimation Theory, One Sample Estimation Problems
- 6 Two Sample Estimation Problems
- 7 Applications
- 8 Midterm 1
- 9 Hypothesis Tests, One- and Two-Sample Tests of Hypothesis
- 10 Applications
- 11 Analysis of Variance, Regression and Correlation
- 12 Regression and Correlation
- 13 Midterm 2
- 14 Chi-squared Test, Non-parametric Tests
- 15 Final Exam